increasingly being integrated into oncology practice in India, although its impact varies, depending on the type and stage of cancer," says Dr Randeep Singh, director, Oncology Services, Narayana Hospital, Gurugram.

Genomic sequencing is influencing treatment in advanced lung cancer, bile duct cancer, advanced prostate cancer, breast cancer after multiple lines of therapy and metastatic colon cancer.

SMARTER MEDICINES

Besides cancer, precision medicine is increasingly being used to treat infectious diseases and rare genetic disorders. Tuberculosis, for one. Success in TB treatment depends on early detection. "Early detection allows clinicians to start the correct treatment earlier and prevent the spread of resistant strains," says Dr Gunisha Pasricha, associate director, Infectious Disease Genomics at the genetic testing lab MedGenome. TB culture tests can take three–eight weeks to detect drug resistance. Molecular and genomic tests, such as GeneXpert/CBNAAT, TrueNat and genome sequencing, can identify it within hours to a few days. That difference in time can change treatment outcomes. If needed, whole genome sequencing can detect mutations linked to resistance to 18 tuberculosis drugs, helping doctors to home in on the right medicine.

Thus, the main advantage of precision medicine is accuracy in treatment. Thanks to pharmacogenomics—the study of how an individual's genetic makeup influences her body's response to drugs—patients receive drugs that work for them. It reduces the trial-and-error approach that persists in treating depression, hypertension and diabetes. In cardiology in India, genetic tests for conditions such as familial hypercholesterolemia and inherited cardiomyopathies are now being used to identify high-risk individuals early. In neurology, genetic testing is increasingly used to diagnose and treat Alzheimer's and Parkinson's, epilepsy, spinal muscular atrophy and certain neurodegenerative disorders, where identifying the exact genetic mutation helps doctors choose targeted therapies.

Further, genomics also holds the promise of helping public health systems monitor disease trends, identify outbreak clusters earlier and track drug resistance in real time.



"There are multiple barriers to implementing precision medicine in India.... Cost, infrastructure, lack of trained specialists and limited access to molecular tumour boards are major ones"

DR JYOTI WADHWA
Principal Lead, Medical and Precision Oncology, Apollo Athenaa Women's Cancer Centre, Delhi

THE TALLEST HURDLES

Cost and infrastructure are the biggest barriers to speedier adoption of precision medicine in India. Many advanced genomic tests and targeted therapies are expensive and not covered by insurance. Tumour genome sequencing can cost from Rs 20,000 to Rs 80,000, while targeted cancer therapies can cost from Rs 1.5 lakh to Rs 4 lakh a month. Genome sequencing tests for drug-resistant tuberculosis can cost Rs 10,000 to Rs 25,000 in private facilities, limiting access outside government hospitals. Some gene therapies for rare diseases can cost several crores of rupees, making them inaccessible for most patients.

Cost is only part of the problem. Precision medicine requires specialised laboratories, bioinformatics systems and trained molecular scientists who can interpret genomic reports and guide treatment decisions. Many of these facilities are still concentrated in metropolises with large hospitals, like AIIMS Delhi, Tata Memorial Mumbai and the Narayana and Apollo hospital chains. "There are multiple barriers to implementing precision medicine in India," says Dr Jyoti Wadhwa, principal lead of Medical and Precision Oncology, Apollo Athenaa Women's Cancer Centre, Delhi. "Cost, infrastructure, lack of trained specialists and limited access to molecular tumour boards (panel of medical experts who review complex cancer cases based on genomics) are major barriers."

Another major challenge lies in interpreting data generated from DNA sequencing alongside clinical data, treatment history and patient outcomes. In India, medical records are often unstructured and stored across multiple systems, making it difficult to combine genomic data with clinical information in a meaningful way. "The bigger barriers now lie in standardisation, interpretation of results and availability of trained specialists," says Dr Singh. Genome sequencing produces massive amounts of information and these are increasingly processed through artificial intelligence and bioinformatics tools that turn complex results into simplified clinical reports.

The future of precision medicine in India lies not just in sequencing more genomes, but also in making that information usable. In the next big shift, genomic data will be integrated into everyday clinical decision-making. If that happens, precision medicine could truly transform diagnosis, treatment selection and disease prevention over the next decade. ■